

# Intercepts

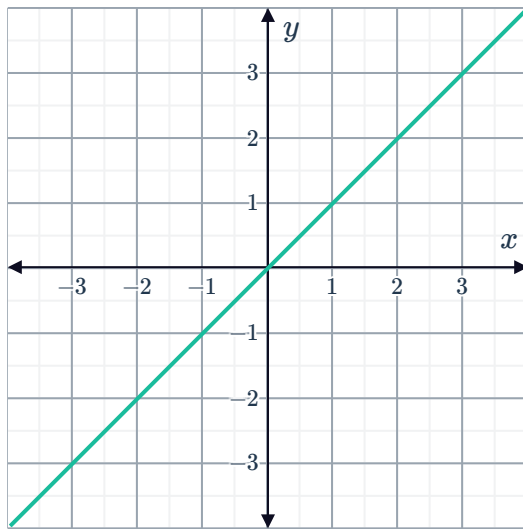


1 For each of the following graphs:

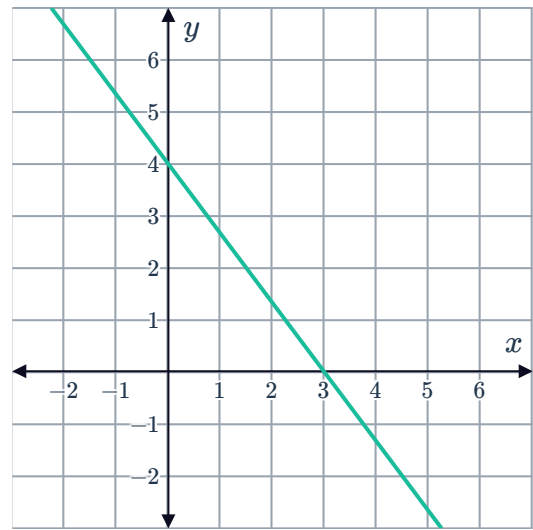
i State the value of the  $x$ -intercept.

ii State the value of the  $y$ -intercept.

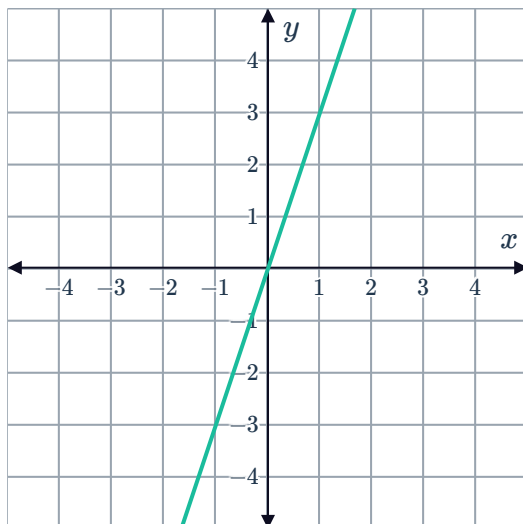
a



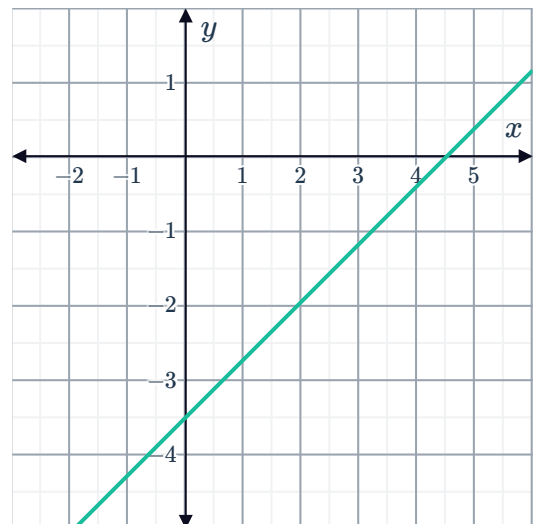
b



c

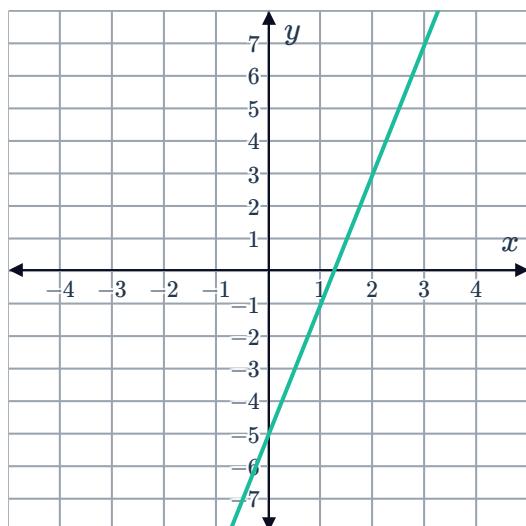


d

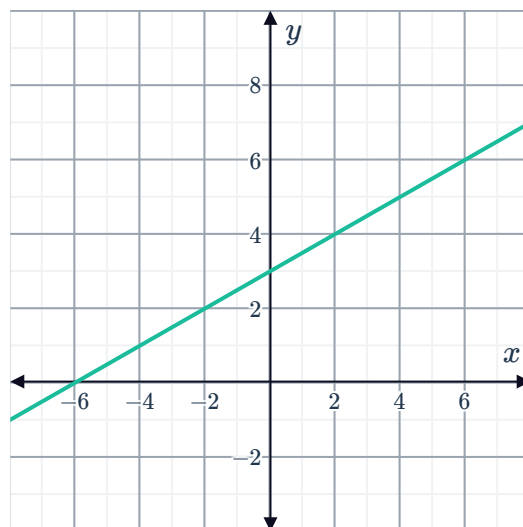


2 Given each linear equation and its graph, state the coordinates of the  $y$ -intercept:

a  $y = 4x - 5$

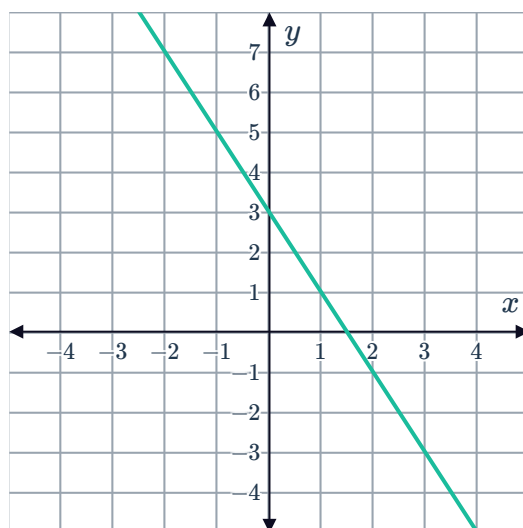


b  $y = \frac{x}{2} + 3$



3 Consider the following graph of the line  $y = -2x + 3$ :

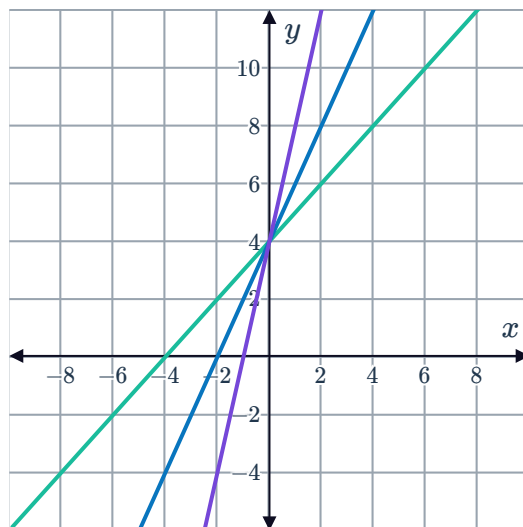
- State the  $y$ -value, when  $x$  is 0.
- Explain the relationship between the value of the  $y$ -intercept and the equation of the line.
- If the equation of a line is  $y = mx + c$ , state the value of the  $y$ -intercept.



4 Consider the following three linear equations and their corresponding graphs:

$$y = x + 4, y = 2x + 4, y = 4x + 4$$

- What do all of the equations have in common?
- What do all of the graphs have in common?
- What conclusion can be made about lines that have the form  $y = mx + 4$ ?



5 Find the value of the  $y$ -intercept of the following lines:



a  $y = 7x + 3$

b  $y = 3x - 5$

c  $y = -8x - 3$

d  $y = -8x + 4$

e  $y = 9x$

f  $y = 2$

g  $y = 2x + \frac{2}{3}$

h  $y = \frac{3x + 8}{5}$

6 Determine whether the following equations represent lines that will cross the  $y$ -axis at 2:

a  $y = 5x + 2$

b  $y = 2 - x$

c  $y = 2x$

d  $y = 2x - 4$

e  $y = x + 2$

f  $y = x - 2$

g  $y = 2$

h  $y = \frac{x + 4}{2}$

7 The  $x$ -intercept occurs when  $y = 0$ . Find the value of the  $x$ -intercept for the following lines:

a  $y = 2x - 2$

b  $3x + y = -3$

c  $y = 4x - 8$

d  $2y + x = -3$

e  $y = 9x$

f  $2y + 2x = 4$

g  $3x - 5y = 1$

h  $x = \frac{3y + 10}{5}$

8 For each of the following equations:

ii Find the coordinates of the  $y$ -intercept.

i Find the coordinates of the  $x$ -intercept.

iii Use the intercepts to sketch the graph of the line.

a  $y = 2x - 4$

b  $y = -2x + 2$

c  $y = 3x - 3$

d  $y = -4x + 8$

---

## Lines through the origin

9 Consider the linear equation  $y = 5x$ .

a Find the coordinates of the  $y$ -intercept.

b Find the coordinates of the  $x$ -intercept.

c Find the value of  $y$  when  $x = 2$ .

d Hence, sketch the graph of the line.

10 Consider the linear equation  $y = -\frac{5x}{4}$ .

a Find the coordinates of the  $y$ -intercept.

b Find the coordinates of the point on the line where  $x = 4$ .

c Hence, sketch the graph of the line.

11 If a line has equation  $y = mx + c$ , explain how you can tell if the line will pass through the



**12** Determine whether the following equations represent lines that will pass through the origin:

**a**  $y = 8x - 8$

**b**  $y = \frac{x}{8}$

**c**  $y = -6x$

**d**  $y = 8x$

**e**  $y = \frac{x}{8}$

**f**  $y = 0$

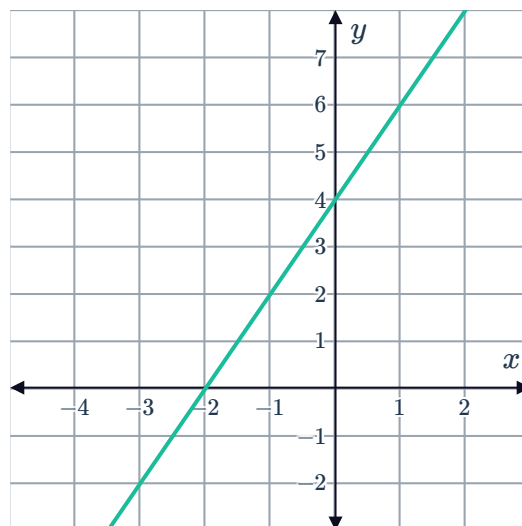
**g**  $y = -x$

**h**  $y = -6x - 6$

## Gradient

**13** Consider the line graph shown:

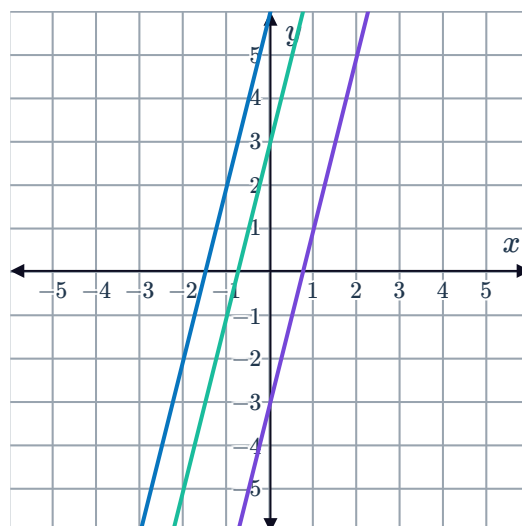
- a** State the  $y$ -value when  $x = 0$ .
- b** State the  $y$ -value when  $x = 1$ .
- c** When the  $x$ -value increases by 1, by how much does the  $y$ -value change?
- d** Hence state the gradient of the line,  $m$ .
- e** The equation of this line is  $y = 2x + 4$ . Explain how to find the gradient from the equation of the line.



**14** Consider the following three linear equations and their corresponding graphs:

$$y = 4x + 3, y = 4x + 6, y = 4x - 3$$

- a** What do all of the equations have in common?
- b** What do all of the graphs have in common?
- c** What conclusion can be made about lines that have the form  $y = 4x + c$ ?



**15** Find the gradient,  $m$ , of the following linear equations:

**a**  $y = 9x + 3$

**b**  $y = -7x + 5$

**c**  $y = \frac{5x}{4} + 2$

**d**  $y = -x + 5$

16 From the following list of equations, select lines that have the same gradient:



- $y = 2 + 7x$
- $y = \frac{x}{7} + 2$
- $y = 5 - 7x$
- $y = 7x$
- $y = 5x + 7$
- $y = 7x - 2$

17 For each linear equation:

i Find the value,  $m$ , of the gradient.

ii Find the value,  $c$ , of the  $y$ -intercept.

a  $y = 2x + 9$

b  $y = 5x - 6$

c  $y = -5x + 8$

d  $y = -4x - 2$

e  $2y = 8x - 1$

f  $3y = -6x - 2$

g  $y - 5x = 4$

h  $2y - 3x = 6$

18 Given the values of  $m$  and  $c$ , write the equation of the line:

a  $m = 2, c = 5$

b  $m = -3, c = 2$

c  $m = -2, c = -1$

d  $m = 4, c = 0$

e  $m = 0, c = -7$

f  $m = 0, c = 4$

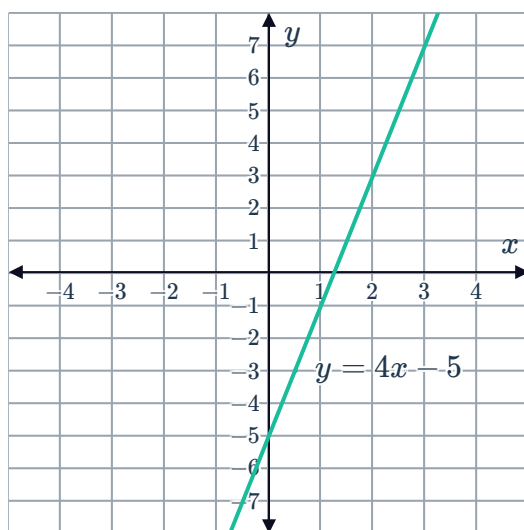
g  $m = \frac{1}{2}, c = -2$

h  $m = -\frac{3}{4}, c = \frac{1}{2}$

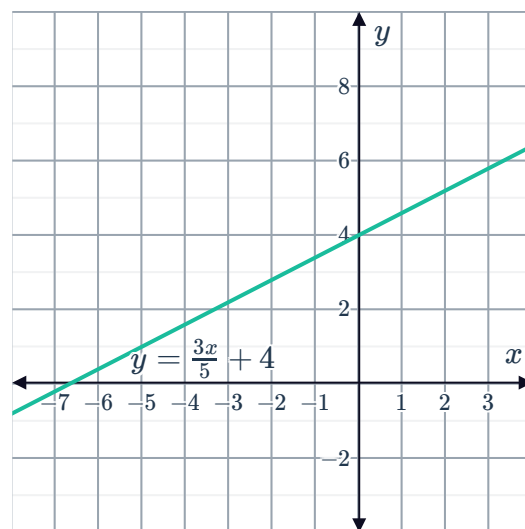
## Approximate solutions of equations

19 Approximate the solutions of the following equations using the graphs provided. Round your answers to one decimal place.

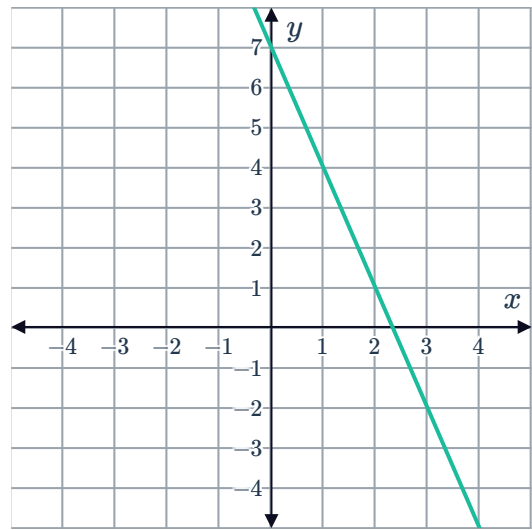
a  $4x - 5 = 0$



b  $\frac{3x}{5} + 4 = 0$



-  **20** Consider the following graph of the line  $y = -3x + 7$ :



- a** Estimate the  $x$ -intercept to one decimal place.
  - b** Hence, estimate the solution to the equation  $-3x + 7 = 0$  to one decimal place.
  - c** Estimate the  $x$ -coordinate of the point on the line where  $y = 6$ .
  - d** Hence, estimate the solution to the equation  $-3x + 7 = 6$  to one decimal place.
- 21** Estimate the solutions of the following equations by first drawing the appropriate graph. Round your answers to one decimal place.

**a**  $5x - 11 = 0$

**b**  $-4x + 9 = 0$

**c**  $6x - 9 = 0$

**d**  $\frac{9x}{5} - 1 = 0$

**e**  $-\frac{7x}{2} + 10 = 0$

**f**  $\frac{3x}{8} - 5 = 0$

**g**  $4x + \frac{1}{2} = 0$

**h**  $\frac{2x}{3} + 7 = 0$